

**1. O/N/23/11/Q36**

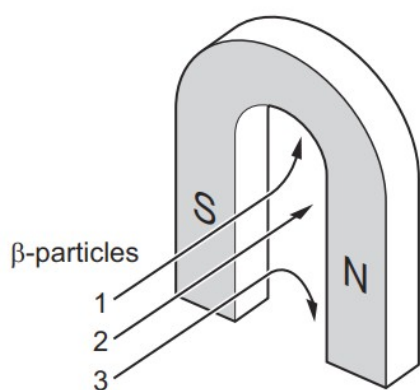
Alpha ( $\alpha$ ) particles and beta ( $\beta$ ) particles are emitted during radioactive decay.

Which particle has the weaker ionising effect and which has the weaker penetrating power?

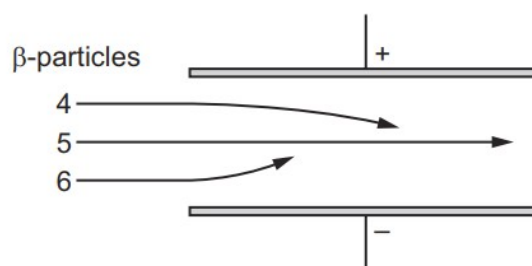
|          | weaker ionising effect | weaker penetrating power |
|----------|------------------------|--------------------------|
| <b>A</b> | $\alpha$               | $\alpha$                 |
| <b>B</b> | $\alpha$               | $\beta$                  |
| <b>C</b> | $\beta$                | $\alpha$                 |
| <b>D</b> | $\beta$                | $\beta$                  |

**2. O/N/23/11/Q37**

A beam of beta ( $\beta$ ) particles passes into a magnetic field and into an electric field.



into magnetic field



into electric field

Which numbers show the paths of the beam through the fields?

- A** 1 and 4      **B** 1 and 6      **C** 2 and 5      **D** 3 and 6

**3. O/N/23/11/Q38**

The isotope radon-222 has a half-life of 3.8 days.

Which statement about a sample containing 1.0 mg of radon-222 is correct?

- A** After 1.9 days there is 0.50 mg of radon-222 left.  
**B** After 3.8 days there is 0.50 mg of radon-222 left.  
**C** Its half-life is half that of a 2.0 mg sample.  
**D** There is no radon-222 left after 7.6 days.

**4. O/N/23/12/Q39**

The half-life of a radioactive isotope is 2.5 minutes.

This isotope is the only radioactive isotope in a radioactive sample.

Between time  $t = 0$  and  $t = 7.5$  minutes, the mass of this isotope in the sample decreases by  $70\text{ }\mu\text{g}$ .

What is the mass of this isotope at  $t = 0$ ?

- A**  $75\text{ }\mu\text{g}$       **B**  $80\text{ }\mu\text{g}$       **C**  $210\text{ }\mu\text{g}$       **D**  $560\text{ }\mu\text{g}$

**5. M/J/23/12/Q36**

A sample of a radioactive isotope with a very long half-life is placed next to a detector. Sheets of different materials are inserted, one at a time, between the radioactive sample and the detector.

The background count rate is subtracted from the readings on the detector and the corrected count rates are recorded in the table.

| type of material     | <u>corrected count rate</u><br>counts/s |
|----------------------|-----------------------------------------|
| no sheet inserted    | 450                                     |
| thin aluminium sheet | 381                                     |
| thin lead sheet      | 285                                     |
| thin paper sheet     | 452                                     |

Which types of radiation are emitted by the radioactive sample?

- A** alpha-particles and beta-particles only  
**B** alpha-particles only  
**C** beta-particles and gamma radiation only  
**D** gamma radiation only

**6. M/J/23/12/Q37**

In a fission reactor, which particle causes a uranium-235 nucleus to split?

- A** alpha-particle  
**B** gamma ray  
**C** neutron  
**D** proton

**7. M/J/23/12/Q38**

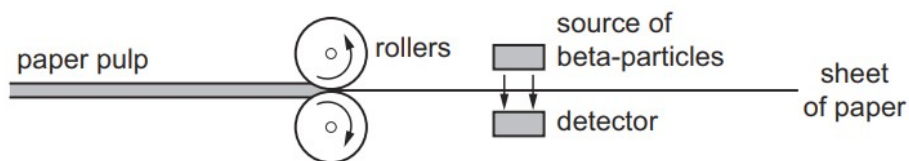
What is the purpose of moderators in nuclear reactors?

- A** to absorb all the neutrons in order to stop the chain reaction  
**B** to produce more neutrons in order to create a chain reaction  
**C** to remove thermal energy in order to control the chain reaction  
**D** to slow down neutrons in order to continue the chain reaction

**8. M/J/23/11/Q37**

The diagram shows how the thickness of paper is measured during manufacture.

If the sheet is too thick, fewer beta-particles can reach the detector.



Why are beta-particles used for this purpose?

- A** Alpha-particles are all stopped by the paper.
- B** Alpha-particles make the paper radioactive.
- C** Gamma radiation is all stopped by the paper.
- D** Gamma radiation is only produced by sources with a short half-life.

**9. M/J/23/11/Q38**

A sample of tritium has a mass of 32 mg.

Tritium has a half-life of 12 years.

How much of the sample of tritium decays in 36 years?

- A** 4.0 mg      **B** 8.0 mg      **C** 24 mg      **D** 28 mg

**10. O/N/22/11/Q36**

Three types of radiation emitted by unstable nuclei are helium nuclei, short wavelength electromagnetic waves and electrons.

What are these three types of radiation?

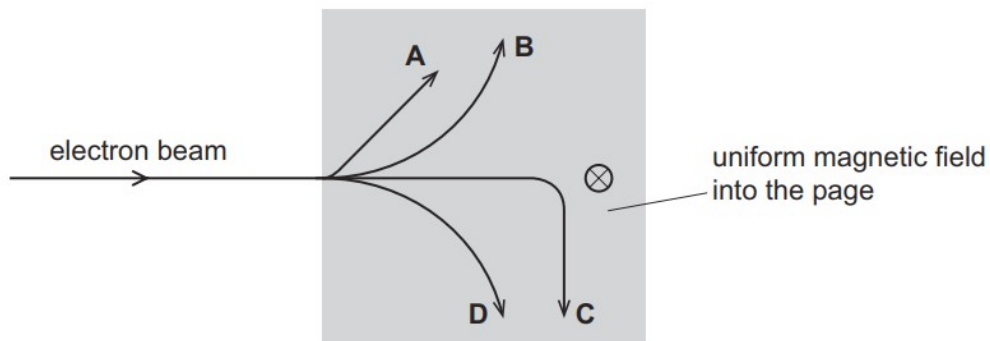
|          | helium nuclei | electromagnetic waves | electrons |
|----------|---------------|-----------------------|-----------|
| <b>A</b> | alpha         | gamma                 | beta      |
| <b>B</b> | beta          | gamma                 | alpha     |
| <b>C</b> | gamma         | alpha                 | beta      |
| <b>D</b> | gamma         | beta                  | alpha     |

**11. O/N/22/11/Q37**

A beam of electrons is fired into a uniform magnetic field as shown.

The direction of the magnetic field is into the page.

Which path do the electrons follow?



**12. O/N/22/11/Q38**

Which statement about all three types of radioactive emissions is correct?

- A** They are all completely absorbed by a thin aluminium sheet.
- B** They are all deflected by electric fields.
- C** They are all deflected by magnetic fields.
- D** They all cause ionisation of gases.

**13. O/N/22/11/Q39**

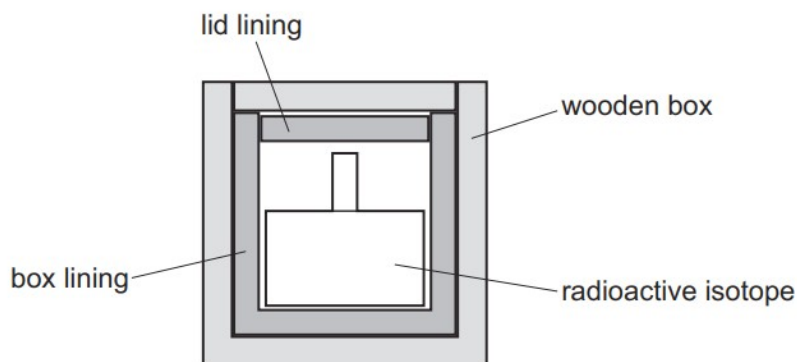
What is meant by the half-life of a radioactive substance?

- A** half the time for all the unstable nuclei to decay
- B** half the time it takes an unstable nucleus to emit radiation
- C** the time for half the unstable nuclei to decay
- D** the time for the nucleon number to halve



**14. O/N/22/11/Q40**

Radioactive isotopes that emit alpha-particles, beta-particles and gamma rays are stored in wooden boxes that are lined with a material that prevents most radiation escaping from the box.



Which material is used for the lining?

- A** copper
- B** glass
- C** lead
- D** plastic

**15. O/N/22/12/Q39**

Different radioactive emissions have different characteristics.

A student lists the following characteristics for an alpha-particle.

characteristic P It has a positive charge.

characteristic Q It has a high penetrating ability.

characteristic R It is a component of the electromagnetic spectrum.

Which characteristics are correct?

- A** P only
- B** R only
- C** P and Q
- D** Q and R

**16. O/N/22/12/Q40**

A wooden object is believed to be 6000 years old.

Which isotope, found in the object, is used to determine the age of the object?

- A**  $^{12}\text{C}$
- B**  $^{13}\text{C}$
- C**  $^{14}\text{C}$
- D**  $^{15}\text{C}$

**17. O/N/21/11/Q38**

Technetium-99m is a radioactive isotope used in medical scanning. It is injected into the body and its emissions are detected outside the body.

Which characteristics of technetium-99m make it suitable for use in medical scanning?

- A It has a long half-life and emits alpha radiation.
- B It has a long half-life and emits gamma radiation.
- C It has a short half-life and emits alpha radiation.
- D It has a short half-life and emits gamma radiation.

**18. O/N/21/11/Q39**

Which statement about the production of electricity in a nuclear power station is correct?

- A In the reactor, the main reaction occurs when protons hit uranium nuclei.
- B The process taking place in the reactor is called nuclear fusion.
- C The reactor produces energy to boil water and to produce steam.
- D Carbon dioxide is the major waste product from the reactor.

**19. O/N/21/12/Q38**

A radioactive sample contains an isotope that emits alpha particles.

Which quantity stays constant?

- A the half-life of the isotope
- B the mass of the sample
- C the number of neutrons in the sample
- D the rate of decay of the isotope

**20. M/J/21/12/Q39**

What does an alpha-particle consist of?

- A two electrons and four neutrons only
- B two protons and two neutrons only
- C two protons and four neutrons only
- D two protons, two electrons and two neutrons

**21. M/J/21/12/Q40**

The count rate from a radioactive source falls from 4000 counts per minute to 500 counts per minute in 72 minutes.

What is the half-life of the source?

- A 8 minutes
- B 9 minutes
- C 18 minutes
- D 24 minutes

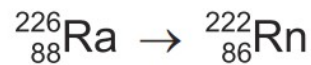
**22. M/J/21/11/Q37**

Which statement about nuclear fusion is correct?

- A Nuclear fusion occurs at low temperatures.
- B Nuclear fusion occurs only between heavy nuclei.
- C Nuclear fusion occurs in the formation of many stars.
- D Nuclear fusion powers most electricity-generating stations.

**23. M/J/21/11/Q38**

In one radioactive decay, radium-226 decays to radon-222 as shown.



Which particles are also produced?

- A both an alpha-particle and a beta-particle
- B an alpha-particle only
- C a beta-particle only
- D a neutron

**24. O/N/20/11/Q40**

An isotope P is radioactive and has a half-life of 7.0 years. A sample initially contains 0.016 kg of P.

After how long will the sample contain 0.0020 kg of P?

- A 7.0 years      B 14 years      C 21 years      D 28 years

**25. O/N/20/12/Q39**

What is the definition of the half-life of a radioactive element?

- A half of the time it takes for one atom to decay
- B half of the time taken for all the atoms in a sample to decay
- C the time taken for half of the atoms in a sample to decay
- D the time taken for the nucleon number (mass number) to halve

**26. M/J/20/11/Q38**

Radioactivity is used in several activities that take place in hospitals.

Which activity **never** uses radioactivity?

- A cooking meals
- B diagnosing illnesses
- C sterilising equipment
- D treating illnesses

**27. M/J/20/11/Q39**

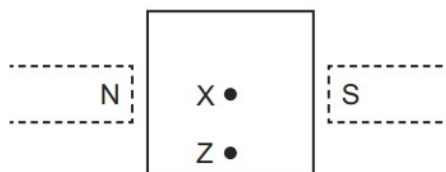
$^{14}_6\text{C}$  is a natural radioactive isotope of carbon and is present in all living things. It has a half-life of 5700 years.

How old is a bone that is found to have only  $\frac{1}{4}$  of the natural proportion of  $^{14}_6\text{C}$ ?

- A 5700 years
- B 11 400 years
- C 17 100 years
- D 22 800 years

**28. O/N/19/11/Q40**

A narrow beam of radiation from a radioactive source travels out of the page through the point X.



The poles of a magnet are placed on either side of the beam. Radiation is now detected both at X and at Z.

Which types of radiation are present in the beam?

- A alpha and beta radiation
- B alpha and gamma radiation
- C alpha, beta and gamma radiation
- D beta and gamma radiation



**29. O/N/19/12/Q39**

Radioactive substances are used as tracers in medical examinations.

A liquid tracer is injected into a patient. A detector 20 cm above the patient shows where the radioactive substance is in the body.

What is the best radioactive substance to use?

- A** an alpha-particle emitter with a half-life of a few hours
- B** an alpha-particle emitter with a half-life of a few years
- C** a gamma ray emitter with a half-life of a few hours
- D** a gamma ray emitter with a half-life of a few years

**30. O/N/19/12/Q40**

An isotope of lead,  $^{212}_{82}\text{Pb}$ , is radioactive. After three decays, two beta-particles and an alpha-particle have been emitted. Another radioactive isotope of lead is formed.

What is the mass number (nucleon number) of the new isotope of lead?

- A** 200                      **B** 208                      **C** 212                      **D** 216

**31. M/J/19/12/Q40**

A proton has charge  $q$  and mass  $m$ .

What is the charge and what is the mass of an alpha-particle?

|          | charge | mass |
|----------|--------|------|
| <b>A</b> | $q$    | $2m$ |
| <b>B</b> | $2q$   | $2m$ |
| <b>C</b> | $2q$   | $4m$ |
| <b>D</b> | $4q$   | $2m$ |

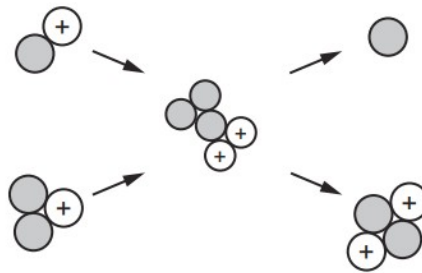
**32. O/N/18/11/Q39**

What does the alpha-radiation given off by radioactive nuclei consist of?

- A** fast-moving protons
- B** helium nuclei
- C** microwaves
- D** radio waves

**33. O/N/18/11/Q37**

The diagram shows a nuclear reaction.



The shaded particles are uncharged.

What are the products shown in this reaction?

- A** a helium atom and a neutron
- B** a helium atom and a proton
- C** a helium nucleus and a neutron
- D** a helium nucleus and a proton

**34. O/N/18/12/Q38**

Which statement about the reactor in a nuclear power station is correct?

- A** In the reactor, the main reaction occurs when protons hit uranium nuclei.
- B** The process taking place in the reactor is called nuclear fusion.
- C** The reactor produces energy to boil water and to produce steam.
- D** Carbon dioxide is the major waste product from the reactor.

**35. M/J/18/11/Q38**

A radioactive isotope of hydrogen has a half-life of 12 years.

A sample contains 40 million atoms of this isotope.

How many atoms of this isotope are left after 24 years?

- A** 20 million
- B** 10 million
- C** 5 million
- D** zero

**36. M/J/18/11/Q37**

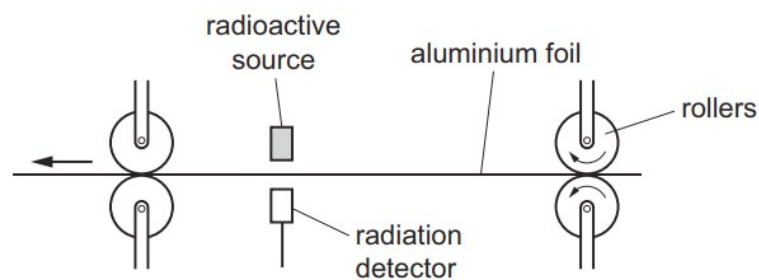
Which of alpha, beta and gamma radiations are waves?

- A** alpha only
- B** beta only
- C** gamma only
- D** alpha and beta

**37. M/J/18/11/Q36**

Thin aluminium foil is used in cooking.

The diagram shows a system used to control the thickness of aluminium foil being made in a factory.



What is the most suitable radioactive source for the system?

|          | source | half-life of source |
|----------|--------|---------------------|
| <b>A</b> | beta   | 1 hour              |
| <b>B</b> | beta   | 6 years             |
| <b>C</b> | gamma  | 1 hour              |
| <b>D</b> | gamma  | 6 years             |

**38. O/N/17/11/Q40**

Which types of radiation may be emitted by radioactive decay?

- A** alpha-particles and gamma rays
- B** microwaves and infra-red
- C** radio waves and microwaves
- D** ultra-violet and X-rays

**39. O/N/17/12/Q37**

A sample contains atoms of an isotope that has a half-life of 7.2 years.

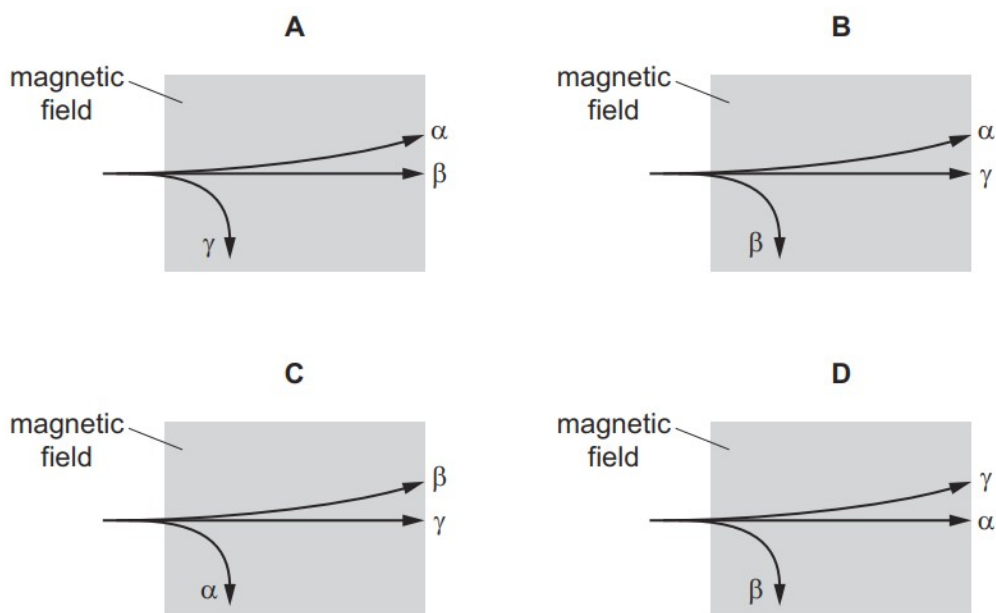
Which quantity halves every 7.2 years?

- A** the mass of the sample
- B** the nucleon (mass) number
- C** the number of atoms of the isotope
- D** the proton (atomic) number

**40. O/N/17/12/Q38**

A beam, consisting of alpha-particles ( $\alpha$ ), beta-particles ( $\beta$ ), and gamma rays ( $\gamma$ ), passes into a magnetic field.

Which diagram shows their paths in the magnetic field?



**41. O/N/17/12/Q40**

A nucleus of phosphorus  $^{32}_{15}\text{P}$  emits a beta-particle to form a new nucleus.

What is the nucleon number and what is the proton number of the new nucleus?

|          | nucleon number<br>(mass number) | proton number<br>(atomic number) |
|----------|---------------------------------|----------------------------------|
| <b>A</b> | 28                              | 13                               |
| <b>B</b> | 31                              | 14                               |
| <b>C</b> | 31                              | 15                               |
| <b>D</b> | 32                              | 16                               |



**42. M/J/17/12/Q39**

Which row is correct for nuclear fission and for nuclear fusion?

|          | fission                 | fusion                             |
|----------|-------------------------|------------------------------------|
| <b>A</b> | produces larger nuclei  | is the energy source of a star     |
| <b>B</b> | produces larger nuclei  | releases energy in a power station |
| <b>C</b> | produces smaller nuclei | is the energy source of a star     |
| <b>D</b> | produces smaller nuclei | releases energy in a power station |

**43. M/J/17/12/Q40**

What happens to the proton number (atomic number) of a nucleus as it emits an alpha-particle?

- A** It decreases by one.
- B** It decreases by two.
- C** It decreases by four.
- D** It does not change.

**44. O/N/16/11/Q38**

Which types of radiation may be emitted by radioactive nuclei?

- A** beta and gamma
- B** microwaves and infra-red
- C** radio waves and microwaves
- D** ultra-violet and X-rays

**45. O/N/16/11/Q39**

A freshly made sample of radioactive material gives a count rate of 8000 counts per minute. After 20 days, it gives a count rate of 500 counts per minute.

What is the half-life of the material?

- A** 4.0 days
- B** 5.0 days
- C** 20 days
- D** 80 days

**46. O/N/16/11/Q40**

Which number is always equal to the proton number of a neutral atom of a radioactive isotope?

- A** the number of electrons in the atom
- B** the number of neutrons in the atom
- C** the number of nucleons in the atom
- D** the number of particles in the atom

**47. O/N/16/12/Q40**

A nucleus of uranium  $^{238}_{92}\text{U}$  decays to thorium by emitting an alpha-particle.

What is the resulting thorium nucleus?

- A**  $^{234}_{90}\text{Th}$       **B**  $^{236}_{90}\text{Th}$       **C**  $^{238}_{93}\text{Th}$       **D**  $^{239}_{92}\text{Th}$

**48. M/J/16/12/Q37**

Which nucleus is produced when americium-241 ( $^{241}_{95}\text{Am}$ ) emits an alpha-particle?

- A**  $^{237}_{93}\text{Np}$       **B**  $^{237}_{97}\text{Bk}$       **C**  $^{245}_{93}\text{Np}$       **D**  $^{245}_{97}\text{Bk}$

**49. M/J/16/12/Q38**

Which states the three types of radiation emitted by radioactive isotopes in order of their ionising effect from highest to lowest?

- A** alpha-particles, beta-particles, gamma-rays  
**B** alpha-particles, gamma-rays, beta-particles  
**C** beta-particles, gamma-rays, alpha-particles  
**D** gamma-rays, beta-particles, alpha-particles

**50. M/J/16/12/Q39**

Which statement about the half-life of a radioactive isotope is correct?

- A** Half-life changes as the isotope decays.  
**B** Half-life is the time it takes for the nucleon number of the isotope to halve.  
**C** Half-life is half the time it takes for the number of nuclei of the isotope to decrease to zero.  
**D** Half-life is the time it takes for the number of nuclei of the isotope to decrease by half.

**51. M/J/16/11/Q38**

Which type of radiation consists of electrons?

- A** alpha-particles  
**B** beta-particles  
**C** gamma-rays  
**D** X-rays

**52. M/J/16/11/Q39**

A sample of a radioactive isotope produces a count rate of 10 000 counts per minute. The half-life of the isotope is one day.

What was the count rate of the sample two days ago?

- A 2500 counts per minute
- B 5000 counts per minute
- C 20 000 counts per minute
- D 40 000 counts per minute

**53. O/N/15/11/Q39**

Which type of electromagnetic radiation is produced during radioactive decay?

- A alpha-particles
- B beta-particles
- C gamma-rays
- D X-rays

**54. O/N/15/11/Q40**

Which row states where nuclear fusion occurs and what nuclear fusion is?

|   | nuclear fusion occurs in | nuclear fusion is             |
|---|--------------------------|-------------------------------|
| A | a power station          | the joining of small nuclei   |
| B | a power station          | the splitting of large nuclei |
| C | a star                   | the joining of small nuclei   |
| D | a star                   | the splitting of large nuclei |

**55. O/N/15/12/Q39**

A radioactive isotope  $^{14}\text{C}$  emits beta-particles.

A sample of this isotope is left for a period of time that is equal to its half-life.

Which two quantities decrease to **half** of their initial value during this time?

- A the decay rate and the number of protons
- B the mass of the sample and the number of  $^{14}\text{C}$  nuclei
- C the number of  $^{14}\text{C}$  nuclei and the decay rate
- D the number of protons and the mass of the sample

**56. M/J/15/12/Q38**

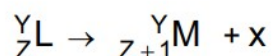
A nucleus of  $^{215}_{84}\text{Po}$  decays by emitting an alpha-particle and the resulting nucleus then decays by emitting a beta-particle.

What is the nucleon number and proton number of the final nucleus?

|          | nucleon number | proton number |
|----------|----------------|---------------|
| <b>A</b> | 211            | 81            |
| <b>B</b> | 211            | 83            |
| <b>C</b> | 212            | 81            |
| <b>D</b> | 212            | 83            |

**57. M/J/15/12/Q39**

A radioactive material decays by this process:



What is particle x?

- A** an electron
- B** a helium nucleus
- C** a neutron
- D** a proton

**58. M/J/15/12/Q40**

The half-life of a radioactive isotope is 24 hours. A sample of this isotope produces an initial count rate of 720 counts per second.

How long does it take for the count rate to fall to 90 counts per second?

- A** 3 hours
- B** 72 hours
- C** 96 hours
- D** 192 hours

**59. M/J/15/11/Q38**

A factory continuously produces plastic sheets. A radioactive isotope and a detector are used to check the thickness of the sheets.

What is the most suitable source to use?

- A** an alpha source with a half-life of a few minutes
- B** an alpha source with a half-life of several years
- C** a beta source with a half-life of a few minutes
- D** a beta source with a half-life of several years



**60. M/J/15/11/Q39**

A source contains a radioactive material.

Without the radioactive source present, a detector records a background count rate of 20 counts per minute.

This source is placed in a fixed position near the detector. Initially a count rate of 520 per minute is recorded.

What count rate is recorded after a time of **two** half-lives of the radioactive source?

- A** 125 counts per minute
- B** 130 counts per minute
- C** 135 counts per minute
- D** 145 counts per minute

**61. M/J/15/11/Q40**

A nucleus contains 94 protons and 240 nucleons. It emits an alpha-particle.

How many protons and how many neutrons are in the nucleus produced?

|          | number of protons | number of neutrons |
|----------|-------------------|--------------------|
| <b>A</b> | 90                | 144                |
| <b>B</b> | 90                | 236                |
| <b>C</b> | 92                | 144                |
| <b>D</b> | 92                | 236                |

**62. O/N/14/11/Q37**

Three types of radiation emitted by unstable nuclei are electrons, helium nuclei and electromagnetic radiation.

What are these types of radiation?

|          | electrons | helium nuclei | electromagnetic radiation |
|----------|-----------|---------------|---------------------------|
| <b>A</b> | alpha     | beta          | gamma                     |
| <b>B</b> | alpha     | gamma         | beta                      |
| <b>C</b> | beta      | alpha         | gamma                     |
| <b>D</b> | beta      | gamma         | alpha                     |



**63. O/N/14/11/Q38**

The energy emitted by the Sun is released when the nuclei of an element fuse together.

Which nuclei, when fusing together, release most of the energy in the Sun?

- A carbon
- B helium
- C hydrogen
- D uranium

**64. O/N/14/11/Q39**

A sample of wood contains  $9.0 \times 10^{16}$  nuclei of carbon-14. The nuclei undergo radioactive decay with a half-life of 5600 years.

How many carbon-14 nuclei remain in this sample after 16 800 years?

- A  $1.1 \times 10^2$       B  $1.1 \times 10^{16}$       C  $3.0 \times 10^{16}$       D  $4.5 \times 10^{16}$

**65. O/N/15/12/Q38**

People working with radioactive materials use a photographic film badge covered in paper. The badge is used to monitor the level of their exposure to radiation.

Which radiation is detected?

- A alpha-particles only
- B beta-particles only
- C gamma-rays and beta-particles
- D gamma-rays only

**66. O/N/14/12/Q39**

When a radioactive atom decays by alpha-particle emission, its nucleus loses

- A 1 proton only.
- B 1 proton and 1 electron.
- C 2 protons and 2 electrons.
- D 2 protons and 2 neutrons.

**67. M/J/14/12/Q39**

Which row states the nature and range of beta-particles in air?

|   | nature                    | range in air |
|---|---------------------------|--------------|
| A | electromagnetic radiation | 1-10 cm      |
| B | electromagnetic radiation | 10-100 cm    |
| C | electron                  | 1-10 cm      |
| D | electron                  | 10-100 cm    |

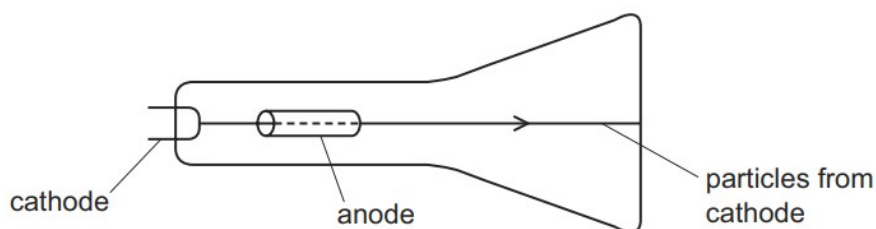
**68. M/J/14/12/Q40**

Which particle has the smallest mass?

- A alpha-particle
- B electron
- C neutron
- D proton

**69. M/J/14/11/Q38**

The diagram shows part of a cathode-ray oscilloscope.



Which particles are emitted by the cathode?

- A electrons
- B helium nuclei
- C neutrons
- D protons

**70. M/J/14/11/Q39**

Which row gives the range and electrical charge of an alpha-particle?

|   | range in air      | electrical charge |
|---|-------------------|-------------------|
| A | a few centimetres | negative          |
| B | a few centimetres | positive          |
| C | a few metres      | negative          |
| D | a few metres      | positive          |

**71. M/J/14/11/Q40**

$^{14}_6\text{C}$  represents a nuclide of the element carbon and  $^{14}_7\text{N}$  a nuclide of nitrogen.

How does a neutral atom of  $^{14}_7\text{N}$  differ from a neutral atom of  $^{14}_6\text{C}$ ?

- A The nitrogen atom has one electron less than the carbon atom.
- B The nitrogen atom has one neutron more than the carbon atom.
- C The nitrogen atom has one proton less than the carbon atom.
- D The nitrogen atom has one proton more than the carbon atom.

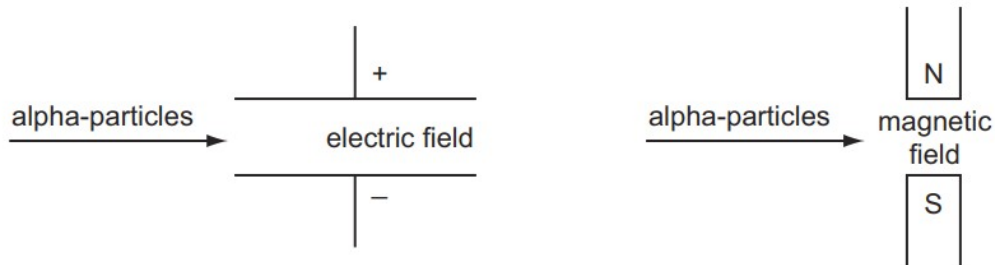
**72. O/N/13/11/Q37**

What are the characteristics of an alpha-particle?

|          | charge   | ionising effect |
|----------|----------|-----------------|
| <b>A</b> | negative | strong          |
| <b>B</b> | negative | weak            |
| <b>C</b> | positive | strong          |
| <b>D</b> | positive | weak            |

**73. O/N/13/11/Q38**

Alpha-particles pass through an electric field or a magnetic field.



How is the path of the particles affected by these fields?

|          | electric field | magnetic field |
|----------|----------------|----------------|
| <b>A</b> | deflected      | deflected      |
| <b>B</b> | deflected      | undeflected    |
| <b>C</b> | undeflected    | deflected      |
| <b>D</b> | undeflected    | undeflected    |

**74. O/N/13/12/Q36**

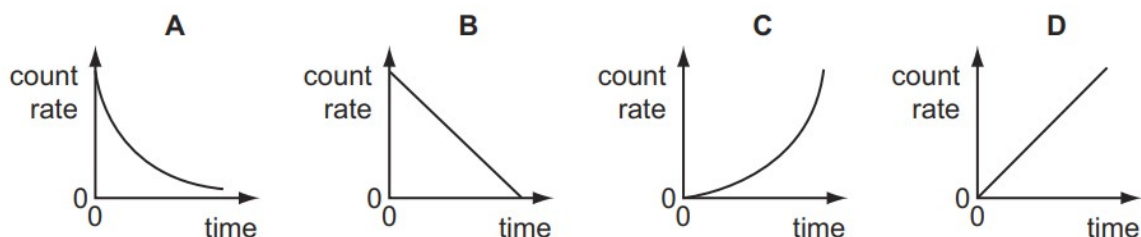
Three types of radiation emitted by unstable nuclei are helium nuclei, short wavelength electromagnetic waves and electrons.

What are these three types of radiation?

|          | helium nuclei | electromagnetic waves | electrons |
|----------|---------------|-----------------------|-----------|
| <b>A</b> | alpha         | gamma                 | beta      |
| <b>B</b> | beta          | gamma                 | alpha     |
| <b>C</b> | gamma         | alpha                 | beta      |
| <b>D</b> | gamma         | beta                  | alpha     |

**75. O/N/13/12/Q38**

Which graph shows how the count rate of a radioactive isotope varies with time?



**76. O/N/13/12/Q39**

For which purpose is the radioactive isotope carbon-14 used?

- A dating of ancient bones
- B fuel for a nuclear power station
- C killing cancerous cells
- D operating mobile telephones

**77. M/J/13/12/Q39**

The count rate from a radioactive source falls from 4000 counts per minute to 500 counts per minute in 72 minutes.

What is the half-life of the source?

- A 8 minutes
- B 9 minutes
- C 18 minutes
- D 24 minutes

**78. M/J/13/12/Q40**

In the Geiger-Marsden experiment, a beam of alpha-particles is fired at a thin sheet of gold in a vacuum. The majority of the alpha-particles pass straight through the sheet without being deflected.

What does this show?

- A The alpha-particle is uncharged.
- B The alpha-particle is very large.
- C The nucleus is positively charged.
- D The nucleus is very small.



**79. M/J/13/11/Q39**

A school keeps radioactive sources for use in radioactivity experiments in a laboratory.

The background radiation is measured at the start of an experiment.

Which statement is correct?

- A** The background radiation is caused by the school's radioactive sources in the laboratory.
- B** The background radiation is present when there are no radioactive sources in the laboratory.
- C** The background radiation is radiation that is not detected in radioactivity experiments.
- D** The background radiation is the same in laboratories in different countries.

**80. O/N/12/11/Q39**

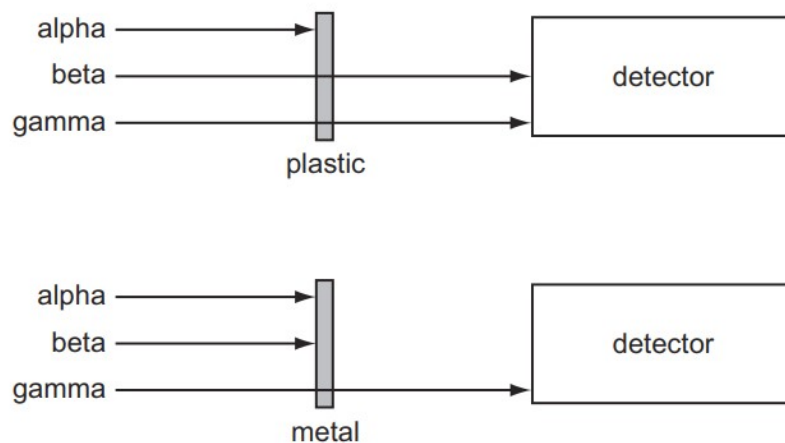
In nuclear .....1....., hydrogen nuclei .....2..... to form helium nuclei, releasing energy.

Which words correctly complete gaps 1 and 2?

|          | 1       | 2             |
|----------|---------|---------------|
| <b>A</b> | fission | join together |
| <b>B</b> | fission | split apart   |
| <b>C</b> | fusion  | join together |
| <b>D</b> | fusion  | split apart   |

**81. O/N/12/12/Q39**

The diagram shows the radioactive emissions that pass through a piece of plastic and a piece of metal of the same thickness.



Which types of radioactive emission can distinguish between the plastic and the metal?

- A** alpha, beta and gamma
- B** alpha only
- C** beta only
- D** gamma only

**82. O/N/12/11/Q40**

Which nucleus is produced when thorium-223 ( $^{223}_{90}\text{Th}$ ) emits an alpha-particle?

- A  $^{219}_{88}\text{Ra}$       B  $^{219}_{92}\text{U}$       C  $^{227}_{88}\text{Ra}$       D  $^{227}_{92}\text{U}$

**83. O/N/12/12/Q40**

The background count rate in a laboratory is 10 counts/min.

The measured count rate from a radioactive source is 410 counts/min, which includes the background.

The half-life of the source is 5 days.

What is the measured count rate after 15 days?

- A 10 counts/min  
B 50 counts/min  
C 60 counts/min  
D 110 counts/min

**84. M/J/12/12/Q39**

What is the safest way to dispose of a large quantity of highly radioactive waste?

- A burning it on a fire  
B burying it in dry rock deep underground  
C pouring it down the drain  
D pumping it into a river

**85. M/J/12/12/Q40**

The count rate from a radioactive material falls from 400 counts per second to 50 counts per second in 12 minutes.

What is its half-life?

- A 8 minutes  
B 6 minutes  
C 4 minutes  
D 3 minutes

**86. M/J/12/11/Q40**

An isotope X is radioactive and has a half-life of 4 years. A sample initially contains 8000 atoms of X.

After how many years will the sample contain 1000 atoms of X?

- A 4      B 8      C 12      D 16

**87. M/J/12/11/Q39**

When dealing with radioactive materials there are possible dangers.

Which statement is correct?

- A** Beta-particles can pass through skin and damage body cells.
- B** Materials that emit only alpha-particles must be kept in thick lead containers.
- C** Radioactive materials are safe to handle after two half-lives.
- D** Sources of gamma radiation are dangerous because they have long half-lives.

**88. O/N/11/11/Q39**

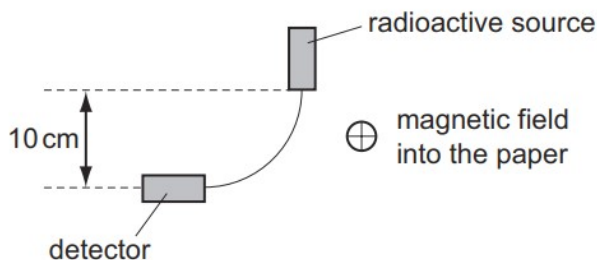
The isotope barium-140 has a half-life of 13 days. A sample of this isotope is kept for 13 days.

Which quantity halves during this time?

- A** the number of atoms of barium-140 in the sample
- B** the number of electrons in an atom of barium-140
- C** the number of nucleons in an atom of barium-140
- D** the number of protons in a nucleus of barium-140

**89. O/N/11/11/Q40**

In a laboratory experiment, particles from a radioactive source are deviated by a magnetic field and reach a detector.



Which particles are deviated and reach the detector?

- A** alpha-particles and gamma-rays
- B** alpha-particles only
- C** beta-particles and gamma-rays
- D** beta-particles only

**90. M/J/11/12/Q38**

A radioactive nuclide  $^{238}_{92}\text{U}$  decays into thorium by emitting an alpha-particle.

The thorium then decays into protactinium by emitting a beta-particle.

What is the symbol for protactinium?

- A**  $^{230}_{90}\text{Pa}$       **B**  $^{234}_{89}\text{Pa}$       **C**  $^{234}_{90}\text{Pa}$       **D**  $^{234}_{91}\text{Pa}$

**91. M/J/11/12/Q40**

Tritium is a radioactive isotope of hydrogen with a half-life of 12 years.

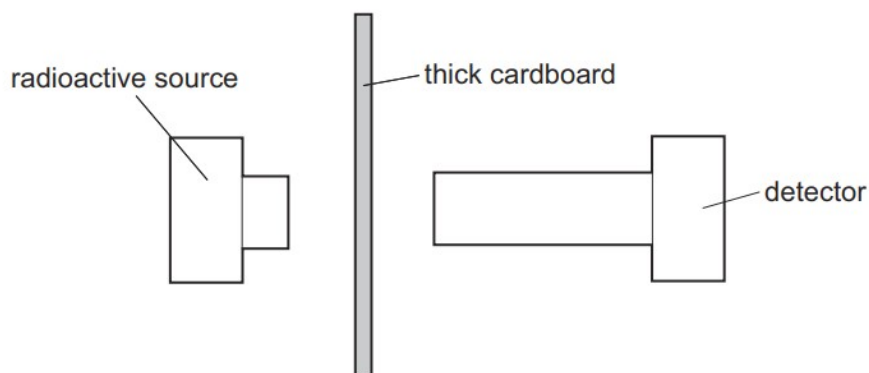
If a sample starts with 40 million atoms of tritium, how many atoms of tritium will be left after 12 years?

- A** 40 million  
**B** 20 million  
**C** 10 million  
**D** 5 million

**92. O/N/10/11/Q38**

A student investigates a radioactive source that emits only alpha-particles. Without any source nearby, the detector shows a low reading.

The source and thick cardboard are placed near the detector, as shown.



What is the reading on the detector now, and why?

|          | detector reading | reason                                                |
|----------|------------------|-------------------------------------------------------|
| <b>A</b> | low              | background radiation is detected                      |
| <b>B</b> | low              | some alpha-particles pass through cardboard           |
| <b>C</b> | zero             | alpha-particles are all absorbed by the cardboard     |
| <b>D</b> | zero             | background radiation is all absorbed by the cardboard |



**93. O/N/10/11/Q40**

The table shows details of two samples of radioactive nuclides X and Y.

| nuclide | number of radioactive atoms at time = 0 | half-life |
|---------|-----------------------------------------|-----------|
| X       | 16 000                                  | 1 day     |
| Y       | 2 000                                   | 2 days    |

After how many days will the number of atoms of nuclide X be equal to the number of atoms of nuclide Y?

- A** 2 days      **B** 4 days      **C** 6 days      **D** 8 days

**94. M/J/10/12/Q40**

When a sample of a radioactive nuclide decays, the count rate falls from 1200 to 150 in three minutes.

What is the half-life of the radioactive nuclide?

- A** 0.75 minutes  
**B** 1.0 minutes  
**C** 3.0 minutes  
**D** 9.0 minutes

**95. O/N/09/Q39**

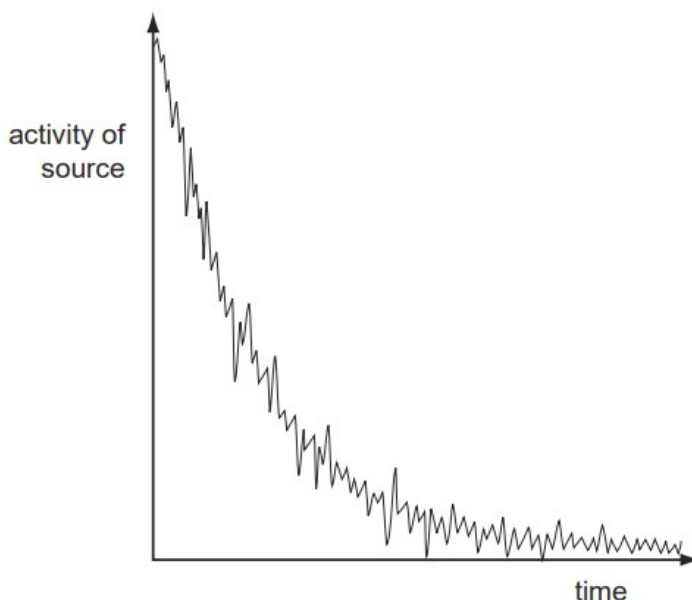
A radioactive isotope has a half-life of 6000 years.

How much time passes before the rate of emission from a sample of this isotope falls to  $\frac{1}{16}$  of its original value?

- A** 6000 years  
**B** 18 000 years  
**C** 24 000 years  
**D** 96 000 years

**96. O/N/09/Q38**

The activity of a radioactive source is measured over a period of time. The graph shows the decay curve.



Why is the curve **not** smooth?

- A** Background radiation has not been subtracted.
- B** Radioactive decay is a random process.
- C** The half-life is not constant.
- D** The temperature is changing.

**97. M/J/09/Q39**

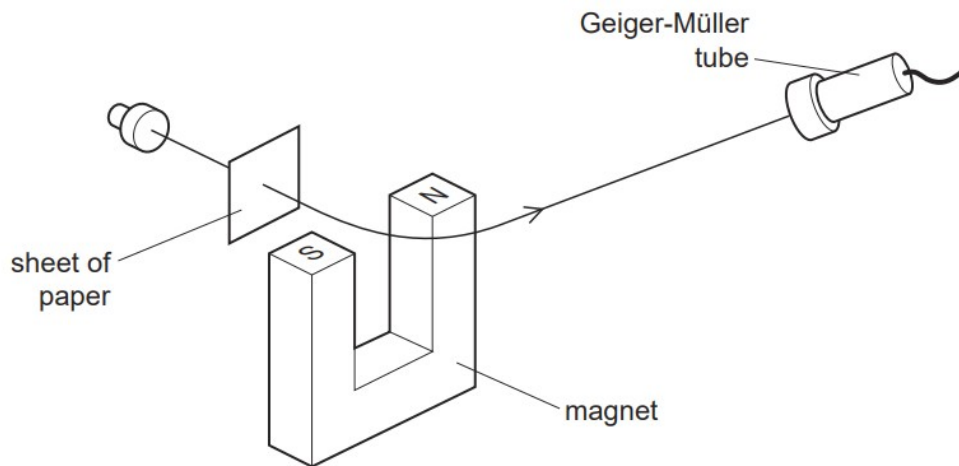
The count-rate from a radioactive source falls from 400 to 50 in 3.0 minutes.

What is the half-life?

- A** 0.75 minutes
- B** 1.0 minutes
- C** 2.7 minutes
- D** 8.0 minutes

**98. M/J/09/Q38**

A radioactive source emits alpha-particles, beta-particles and gamma-rays. A Geiger-Müller tube and counter detect the emissions, which pass through a thin sheet of paper and a strong magnetic field.



What is detected by the Geiger-Müller tube?

- A alpha-particles and beta-particles
- B alpha-particles only
- C beta-particles and gamma-rays
- D beta-particles only

**99. O/N/08/Q38**

What occurs in the decay of a radioactive nucleus?

- A The nucleus absorbs another nucleus.
- B The nucleus absorbs at least one form of radiation.
- C The nucleus always splits into two equal fragments.
- D The nucleus emits at least one form of radiation.

**100. M/J/08/Q38**

Which travels in a straight line across a magnetic field?

- A alpha-particle
- B electron
- C gamma-ray
- D proton

**101. M/J/08/Q39**

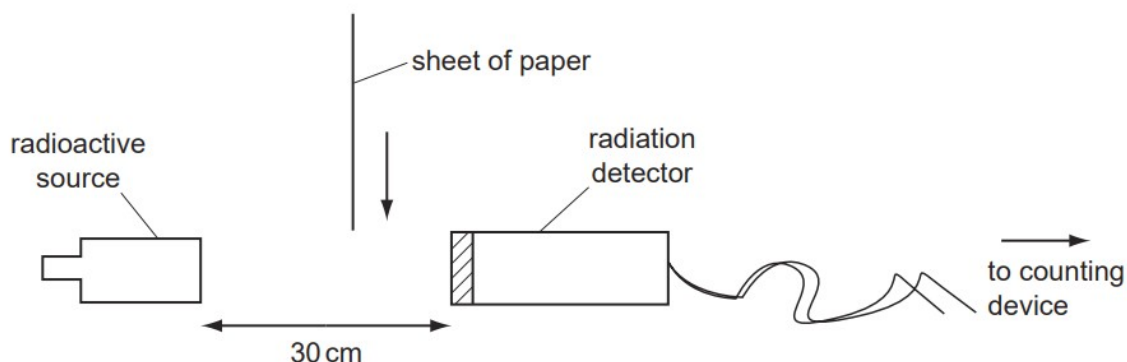
In the treatment of brain cancer, a patient's head is enclosed in a helmet containing a number of radioactive sources. The radiation from each source is directed towards the cancer.

Which nuclide is the most suitable for these sources?

|          | nuclide         | radiation | half-life |
|----------|-----------------|-----------|-----------|
| <b>A</b> | caesium-137     | gamma     | 30 years  |
| <b>B</b> | sodium-24       | beta      | 15 hours  |
| <b>C</b> | strontium-90    | beta      | 29 years  |
| <b>D</b> | californium-246 | alpha     | 36 hours  |

**102. O/N/07/Q38**

An experimenter sets up the following apparatus, in air, to discover whether a radioactive source is emitting alpha-particles.



The experimenter moves the paper between the source and the detector. There is no difference in the count-rate.

Why is this?

- A** Paper does not absorb alpha-particles.
- B** The count-rate due to the background radiation is too high.
- C** The radioactive source is too far from the detector.
- D** The source emits beta-particles and alpha-particles.

**103. O/N/07/Q39**

The half-life of a radioisotope is 2400 years. The activity of a sample is 720 counts/s.

How long will it take for the activity to fall to 90 counts/s?

- A** 300 years      **B** 2400 years      **C** 7200 years      **D** 19 200 years



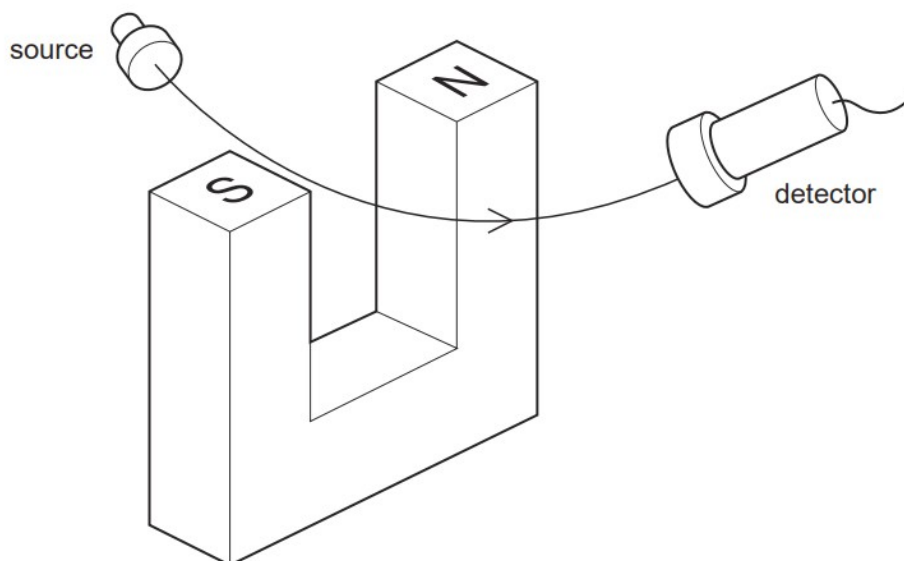
**104. M/J/07/Q38**

Which statement is true for all three types of radioactive emission (alpha-particles, beta-particles and gamma-rays)?

- A They are completely absorbed by a thin aluminium sheet.
- B They are deflected by electric fields.
- C They emit light.
- D They ionise gases.

**105. M/J/07/Q39**

The diagram shows one type of radiation passing between the poles of a strong magnet and being detected.



Which type of radiation is being detected?

- A alpha-particles only
- B beta-particles only
- C gamma-rays only
- D alpha-particles and beta-particles

**106. O/N/06/Q38**

The half-life of a radioactive material is 24 years.

The activity of a sample falls to a fraction of its initial value after 72 years.

What is the fraction?

- A  $\frac{1}{3}$
- B  $\frac{1}{4}$
- C  $\frac{1}{6}$
- D  $\frac{1}{8}$

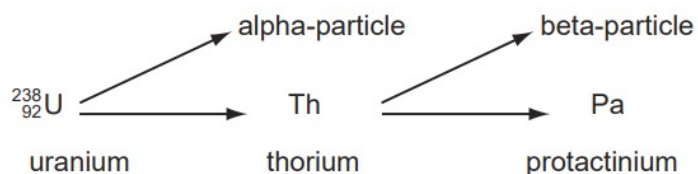
**107. O/N/06/Q39**

Which is the correct comparison of the penetrating power and ionising power of alpha-particles and gamma radiation?

|          | greater penetrating power | greater ionising power |
|----------|---------------------------|------------------------|
| <b>A</b> | alpha                     | alpha                  |
| <b>B</b> | alpha                     | gamma                  |
| <b>C</b> | gamma                     | alpha                  |
| <b>D</b> | gamma                     | gamma                  |

**108. O/N/06/Q40**

The uranium nucleus  ${}_{92}^{238}\text{U}$  emits an alpha-particle to become thorium, which then emits a beta-particle to become protactinium.



What is the proton number (atomic number) of protactinium?

- A** 95                      **B** 91                      **C** 90                      **D** 89

**109. M/J/06/Q39**

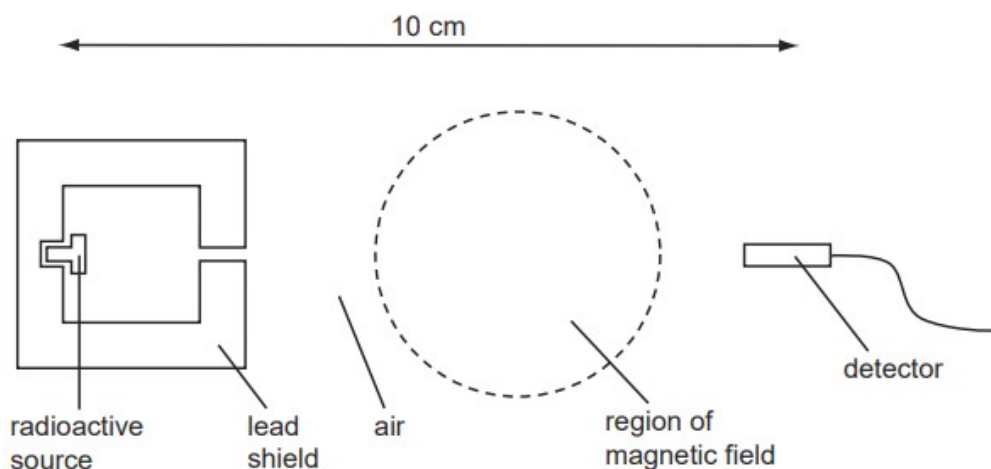
A detector is used to measure the count-rate near a radioactive source. The reading is 4000 counts per minute. After 30 minutes the count-rate has fallen to 500 counts per minute.

What is the half-life of the radioactive source? You may ignore the effects of background radiation.

- A** 3 minutes  
**B** 5 minutes  
**C** 6 minutes  
**D** 10 minutes

**110. M/J/06/Q38**

A student investigates the emission from an unknown radioactive source. The source is 10 cm in front of a detector. A strong magnetic field between the source and the detector is then switched on.



The results are shown.

|                        | average count per minute |
|------------------------|--------------------------|
| without magnetic field | 4500                     |
| with magnetic field    | 2000                     |
| background radiation   | 50                       |

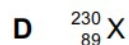
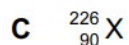
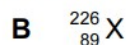
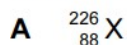
Which radioactive source produced these results?

| source   | emissions from source               |
|----------|-------------------------------------|
| <b>A</b> | alpha-particles and gamma-rays only |
| <b>B</b> | beta-particles only                 |
| <b>C</b> | beta-particles and gamma-rays only  |
| <b>D</b> | gamma-rays only                     |

**111. O/N/05/Q38**

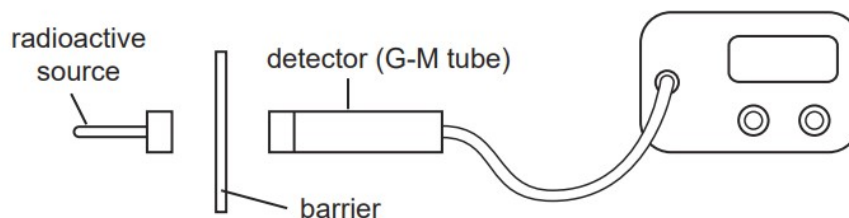
A nucleus is represented by  ${}_{91}^{230}\text{Z}$ . It emits one alpha-particle and then one beta-particle.

What is the resulting nucleus X?



**112. O/N/05/Q37**

The diagram shows the apparatus used in an experiment in which barriers of various materials are placed in turn between different radioactive sources and a detector.



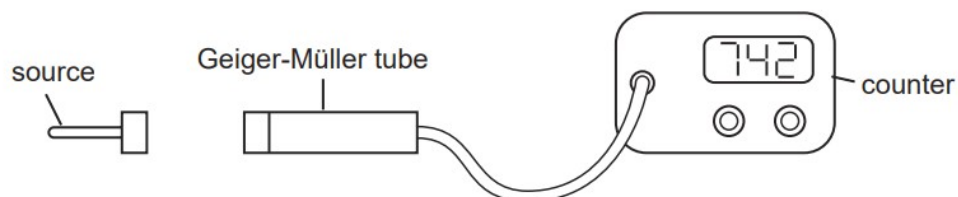
The table shows the count rates recorded by the detector for four sources.

Which source emits alpha-particles only?

| source   | count rate / counts per minute |       |                |            |
|----------|--------------------------------|-------|----------------|------------|
|          | no barrier                     | paper | thin aluminium | thick lead |
| <b>A</b> | 200                            | 200   | 200            | 30         |
| <b>B</b> | 200                            | 30    | 30             | 30         |
| <b>C</b> | 1200                           | 600   | 200            | 30         |
| <b>D</b> | 1200                           | 1200  | 30             | 30         |

**113. M/J/05/Q38**

A radioactive source is placed 3 cm from a Geiger-Müller tube in air. The average count rate is 742 counts / minute.



Then, in three experiments, measurements are taken with sheets of different materials placed between the source and the tube. The results are recorded in the table.

| material of sheet between source and tube | thin card | aluminium foil | thick lead |
|-------------------------------------------|-----------|----------------|------------|
| average count rate / counts per minute    | 273       | 275            | 68         |

Which types of radiation does the source emit?

- A** alpha and beta only
- B** alpha and gamma only
- C** beta and gamma only
- D** alpha, beta and gamma



**114. M/J/05/Q39**

Which equation shows a nuclear fission reaction?

- A**  ${}^2_1\text{H} + {}^2_1\text{H} \rightarrow {}^4_2\text{He}$
- B**  ${}^{14}_7\text{N} + \text{neutron} \rightarrow {}^{15}_7\text{N}$
- C**  ${}^{46}_{21}\text{Sc} \rightarrow {}^{46}_{21}\text{Sc} + \text{gamma ray}$
- D**  ${}^{239}_{92}\text{U} \rightarrow {}^{95}_{38}\text{Sr} + {}^{141}_{54}\text{Xe} + 3 \text{ neutrons}$

**115. O/N/04/Q39**

A nucleus consists of 90 protons and 144 neutrons.

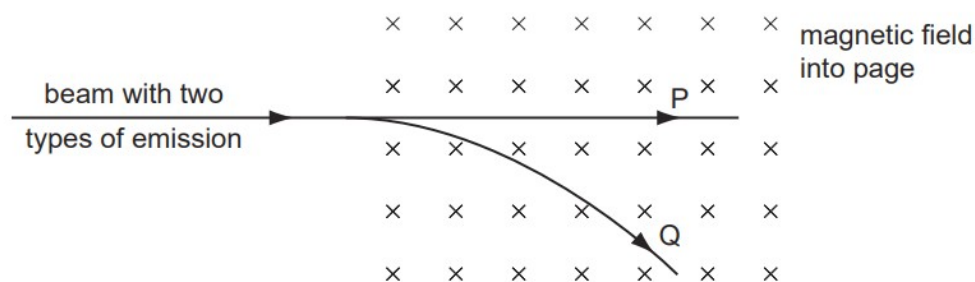
After emitting two beta-particles followed by an alpha-particle, this nucleus has

- A** 86 protons and 140 neutrons.
- B** 86 protons and 142 neutrons.
- C** 90 protons and 140 neutrons.
- D** 90 protons and 142 neutrons.

**116. M/J/04/Q38**

Two types of emission from a radioactive source are separated by passing them through a magnetic field.

The deflections are shown in the diagram.



What are the emissions P and Q?

|          | emission P      | emission Q      |
|----------|-----------------|-----------------|
| <b>A</b> | alpha-particles | gamma-rays      |
| <b>B</b> | beta-particles  | gamma-rays      |
| <b>C</b> | gamma-rays      | alpha-particles |
| <b>D</b> | gamma-rays      | beta-particles  |

117. M/J/04/Q40

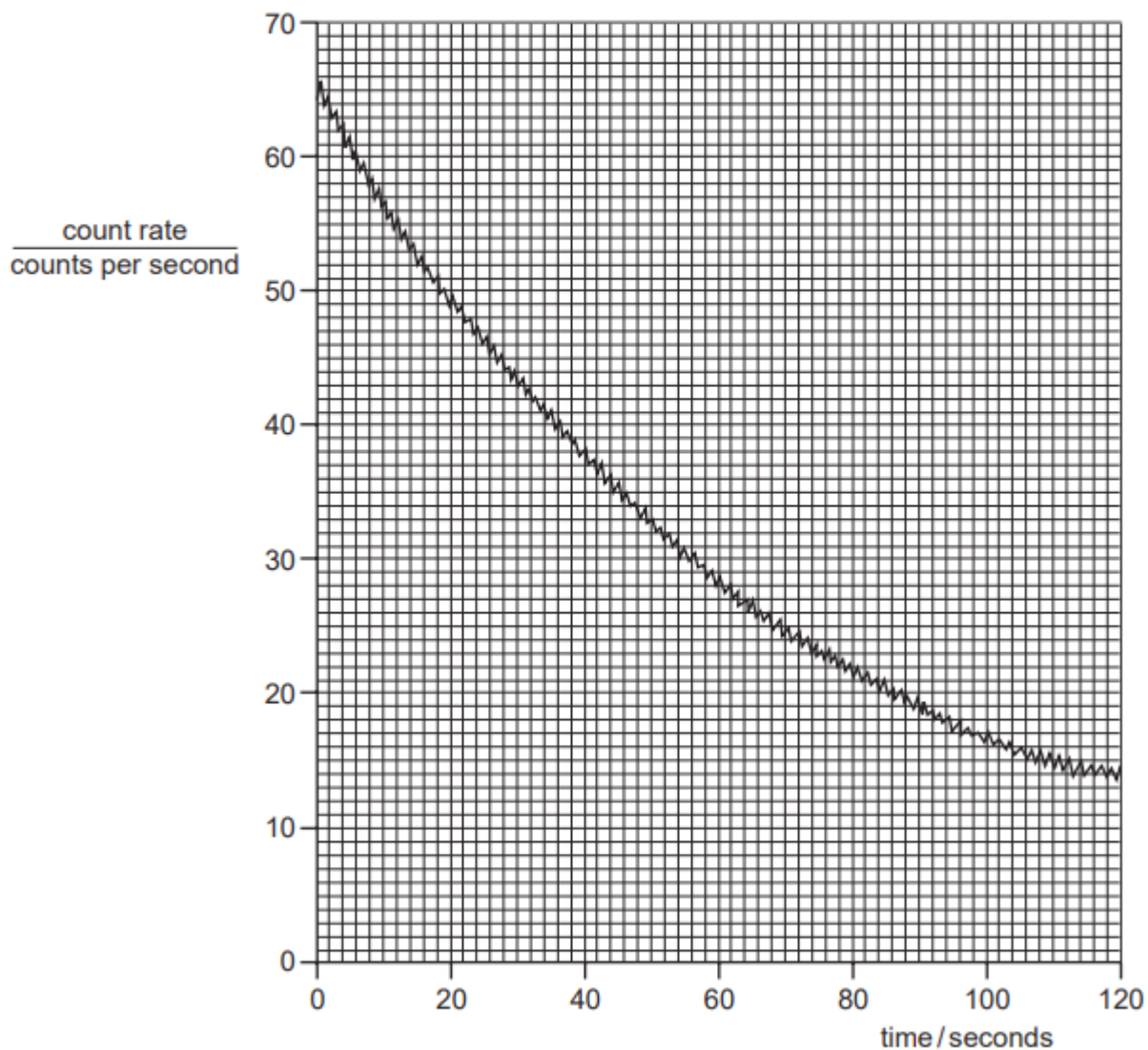
Ra decays with a half-life of 1600 s.

Rn decays with a half-life of 52 s.

Po decays with a half-life of 9.1 s.

Pb decays with a half-life of 10.6 h.

The changing count rate for one of these radioactive nuclides is shown in the graph.



From the half-life shown by the graph, which was the decaying radioactive nuclide?

**A** Ra

**B** Rn

**C** Po

**D** Pb